WEEK 10: NEUROSCIENCE

CRANIAL NERVES AND ITS CONNECTION

THE 12 CN

- Leave the brain and pass through foramina and fissures in skull
- **All** are distributed in neck *except CN 10* - longest

MOTOR NUCLEI OF CN

1.) Somatic motor and branchiomotor nuclei

- Nerve cell and its processes is also known as LMN; like ant gray column of SC
- Function: innervate striated mm
- Impulses: receive from corticobulbar fibers; crosses median plane
- Pathway: pyramidal cells of inf precentral gyrus & adjacent postcentral gyrus -> corona radiata and genu of internal capsule -> midbrain -> synapse either LMN within CN nuclei or through internuncial neurons
  - *1st order neurons* - corticobulbar fibers
  - *2nd order neurons* - internuncial neuron
  - *3rd order neurons* - LMN
- Bilateral connections are present for all CN nuclei except:
  - *Facial nucleus* - lower part of face
  - *Hypoglossal nucleus* - genioglossus mm

2.) General visceral motor nuclei

- Contains somatic and visceral afferent nuclei; like post root ganglion in SC
- Function: parasympathetic innervation of ANS
- Consist: Dorsal motor (10), inf salivatory (9), Sup salivatory (7), lacrimal (7), Edinger-Westphal (3), — 1973

CN 1

- Location: olfactory nerve cell in olfactory mucous membrane; above sup concha

PARTS

1.) Olfactory receptor cells

- Small bipolar nerve cell

2.) Olfactory hairs

- Covers the mucous membrane surface
- Function: react to odors in air and stimulate olfactory cells

3.) Olfactory nerve fibers

- Pass through the cribriform plate of ethmoid bone to enter olfactory bulb
- Unmyelinated; covered with Schwann cells

4.) Olfactory bulb

- Receives axon from contralateral olfactory bulb through olfactory tract
- *Mitral cell* - largest cell
  - *Tufted & granular cells* - small nerve cells that synapse with mitral cell
- *Synaptic glomeruli* - rounded areas formed by the synapse of olfactory fibers with mitral cell

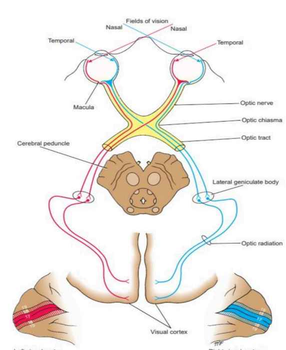
5.) Olfactory tract

- Divided into med & lat olfactory striae before ant perforated substance
  - *Lat* - carries axons to olfactory area of cerebral cortex
    - *Primary olfactory cortex* - periamydaloid and prepiriform area
  - *Med* - carries axons that cross median plane in ant commissure to pass to contralateral olfactory bulb

6.) Entorhinal area — area 28

- Location: parahippocampal gyrus
- Function: appreciation of olfactory sensation
- Also known as *secondary olfactory cortex*

CN 2



- Pathway: axons of ganglionic layer of retina -> optic disc -> exit eye about 3 or 4 mm to nasal side as optic nerve
  - Exit: Optic canal

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- Myelinated fibers; formed by oligodendrocytes due to its similarity to a tract within CNS

PARTS

1.) Optic chiasm

- Location: anterior wall and floor of 3rd ventricle
  - *Anterolateral* - optic nerve
  - *Posterolateral* - optic tract
- Formed by optic nerve of the opposite side
  - Fibers from nasal half of retina crosses the midline and enter optic tract of opposite side
  - Fibers from temporal half pass post in optic tract of same side

2.) Lateral geniculate bodies

- Location: pulvinar of thalamus
- Axons leave to form optic radiation

3.) Optic radiation

- Location: retrolenticular part of internal capsule; terminates in visual cortex (area 17)
- Occupies upper and lower lips of calcarine sulcus
  - *Visual association cortex (area 18, 19)* - recognition of objects and color

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VISUAL REFLEXES

1.) Direct and consensual light reflex

- Constriction of pupil where light is shone
- Contralateral constriction of pupil
  - Pathways: optic nerve, chiasma, tract -> pretectal nucleus -> both side of midbrain -> Edinger-Westphal on both sides -> short ciliary nerves -> eyes -> constrictor pupillae mm of iris

2.) Accommodation light reflex

- Near object contract medial recti
- Lens thickens to increased refractice power; pupil constrict to restrict light waves
  - Pathways: optic nerve, chiasma, tract -> lat geniculate -> optic radiation -> internal capsule -> oculomotor nuclei of midbrain or Edinger-Westphal -> medial recti mm -> short ciliary nerves -> ciliary mm & constrictor pupillae mm of iris

3.) Corneal reflex

- Light on cornea results in blinking of eyelids
  - Pathways: afferent impulse from cornea -> CN V1 -> sensory nucleus -> internuncial neurons -> med longitudinal fasciculus -> motor nucleus -> CN 7 -> blink through orbicularis oculi

4.) Visual body reflex

- Automatic scanning movement of eyes and head when reading
  - Pathways: visual impulses of CN 2 -> optic chiasm, tracts -> sup colliculi -> tectospinal & bulbar tract; ant gray column of SC and motor nuclei

5.) Pupillary skin reflex

- Pupil dilate if skin is pinched
- Believed to have connection with T1-2 SC
  - Pathways: white rami communicantes -> sympathetic trunk -> sup cervical sympathetic ganglion -> internal carotid plexus & long ciliary nerves -> dilator pupillae mm of iris

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NEURONS OF VISUAL PATHWAY AND BINOCULAR VISION

1.) Rods - low-light vision, scotopic vision
2.) Cones - color vision, visual acuity
3.) Bipolar neurons - rods and cones to ganglion cells
4.) Ganglion cells - pass to lat geniculate bodies

IN BINOCULAR VISION

- L & R visual fields -> both retinae -> R visual field vision to nasal of R retina and temporal of L retina -> pass in optic chiasm -> L optic tract
  - ***Lower retinal quadrant (upper visual field)*** - project in lower wall of calcarine
  - ***Upper retinal quadrant*** - upper wall of sulcus
- *Macula lutea* - post part of area 17
- Periphery of retina is represented anteriorly

CN 3

NUCLEI

1.) Main motor nucleus

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- Location: ant gray matter; level of sup colliculus
- Function: supply all extrinsic mm except SO and LR
    - Pathways: nerve fibers -> red nucleus -> midbrain in interpeduncular fossa
- Fibers:
    - *Corticonuclear* - both hemisphere
    - *Tectobulbar* - visual cortex; through sup colliculus
    - *Med longitudinal fasciculus* - CN 4, 6, 8 nuclei

2.) Edinger-Westphal nucleus
- Location: post to main oculomotor nucleus
    - Pathways: parasympathetic neurons -> ciliary ganglion -> postganglionic fibers -> short ciliary nerves -> constrictor pupillae of iris and ciliary mm

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PATHWAYS
- Midbrain -> post cerebral & ant cerebellar arteries -> mid cranial fossa -> divides to sup and inf ramus -> enters orbital cavity through sup orbital fissure

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CN 4
- Supplies SO mm; turns eye downward and laterally

NUCLEI
1.) Trochlear nuclei
- Location: ant gray matter; inf to oculomotor nucleus; level of inf colliculus
- Fibers:
    - *Corticonuclear* - both hemisphere
    - *Tectobulbar* - visual cortex; through sup colliculus
    - *Med longitudinal fasciculus* - CN 4, 6, 8 nuclei

PATHWAYS
- ONLY ONE to leave post brainstem -> midbrain -> decussate contralateral nerve -> mid cranial fossa -> enters orbital cavity through sup orbital fissure

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CN 5
- Largest; sensory part is head contribution & motor is masticatory mm

NUCLEI
1.) Main sensory nuclei
- Location: post pons; lat to motor nucleus

2.) Spinal nuclei
- Location: sup to main sensory; extends inf from medulla to C2 SC

3.) Mesencephalic nuclei
- Location: inf to pons; level of main sensory

4.) Motor nuclei
- Location: pons; med to main sensory

SENSORY COMPONENT
- Pain, temp, touch, pressure, proprioception = trigeminal sensory ganglion
- Half the fibers divide into ascending and descending fibers when they enter pons
    - *Ascending* - terminate in main sensory; touch and pressure
    - *Descending* - terminate in spinal nuclei; pain and temp
- Divisions:
    - *Ophthalmic* - terminate in inf spinal
    - *Maxillary* - terminate in mid spinal
    - *Mandibular* - terminate in sup spinal
- Pathway:
    - Median plane -> trigeminal lemniscus -> terminate on VPM nuclei -> internal capsule -> postcentral gurus (area 3, 1, 2)

MOTOR COMPONENT
- Fibers:
    - *Corticonuclear* - both hemisphere
    - *Reticular formation, red nucleus, tectum,*

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med longitudinal fasciculus
- *Mesencephalic nuclei*

- Supply: masticatory mm, tensor tympani, tensor veli palatini, mylohyoid, ant belly of digastric

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PATHWAYS
- Ant pons -> post cranial fossa -> petrosal part of temporal bone in mid cranial fossa -> Meckel cave (crescent-shaped trigeminal ganglion) -> V1, 2, 3
  - *Ophthalmic* - sensory; exit through supraorbital fissure
  - *Maxillary* - sensory; exit through foramen rotundum
  - *Mandibular* - sensory & motor; exit through foramen ovale
- Little to no gap in dermatomes

CN 6
- Supplies LR mm; turns eye laterally

NUCLEI
1.) Abducens nuclei
- Location: floor of upper 4th ventricle; beneath fasciculus facialis

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PATHWAYS
- Ant pons -> lower pons and medulla -> cavernous sinus -> supraorbital fissure

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CN 7
- Supply:
  - *Motor* - facial expression & auricular mm; stapedius (vibration); post belly of digastric; stylohyoid mm
  - *Sup salivatory* - submandibular & sublingual gland (main); nasal and palatine gland
  - *Lacrimal* - lacrimal gland
  - *Sensory* - ant ⅔ of tongue. floor of mouth, palate
- Lesion: hyperacusis

NUCLEI

1.) Main motor
- Location: reticular formation of lower pons
- Supply: mm of upper face (corticonuclear fibers of both hemisphere), and lower face (corticonuclear fibers from contralateral hemisphere)

2.) Parasympathetic — sup salivatory & lacrimal
- Location: post-lat to main motor
- Fibers:
  - *Sup salivatory* - hypothalamus through descending autonomic pathways; taste sensation from solitary nuclei
  - *Lacrimal* - hypothalamus
- Pathways:
  - Post to med abducens nuclei -> beneath colliculus facialis -> brainstem

3.) Sensory
- Location: upper solitary nuclei; near motor nuclei
- Pathways:
  - Taste -> peripheral axons in geniculate ganglion (nervus intermedius) -> VPM nuclei of contralateral thalamus -> hypothalamic nuclei -> internal capsule & corona radiata -> taste area in lower postcentral gyrus

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PATHWAYS
- Ant brain between pons and medulla -> post cranial fossa with CN 8 -> internal acoustic meatus of petrosal part of temporal bone -> facial canal -> inner ear -> expands to sensory geniculate ganglion -> tympanic cavity -> aditus of mastoid antrum -> pyramid -> stylomastoid foramen

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CN 8
- Internal ear to CNS

1.) Vestibular nerve
- Location: vestibular ganglion in internal acoustic meatus
  - *Utricle* - position of head

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- *Semicircular canal* - movement of head

- Pathways:
  - Ant brainstem (lower pons & upper medulla) -> divides into short ascending and long descending fibers -> small fibers in inf cerebellar peduncles -> cerebellum

VESTIBULAR NUCLEI COMPLEX — 3, 4, 6, 8
- Efferent fibers -> inf cerebellar peduncle -> cerebellum
- Efferent fibers -> lat vestibular nuclei -> SC -> vestibulospinal tract for balance
  - Pass to CN 3, 4, 6 through medial longitudinal fasciculus
- Afferent fibers -> vestibular nuclei -> cerebral cortex -> postcentral gyrus (vestibular area)
- Function: head and eye movement coordination; internal ear information to maintain balance -> influence mm tone of limbs & trunk

2.) Cochlear nerve
- Fibers: located in spiral ganglion of cochlea
- Function: receive sound impulses from organ of Corti
- Pathway:
  - Ant brainstem -> lower pons -> divides into post and ant cochlear nuclei -> cochlear nerve -> trapezoid body & olivary nuclei
  - In midbrain, lat lemniscus fibers either terminate in inf colliculus nuclei or relayed to med geniculate body -> acoustic radiation of internal capsule -> auditory cortex
    - *Primary auditory area (Area 41, 42)* - Heschl's gyrus on sup temporal gyrus
    - *Secondary association area* - recognition and interpretation of sounds from memory

PATHWAY — BOTH
- Ant brain (lower pons & medulla) -> post cranial fossa -> internal acoustic meatus with facial nerve -> different parts of internal ear

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CN 9

NUCLEI

1.) Main motor
- Location: reticular formation of medulla; formed by sup ambiguous nuclei
- Supply: stylopharyngeus mm
- Fibers:
  - *Corticonuclear* - both hemisphere

2.) Parasympathetic — inf salivatory
- Receives information from olfactory system through reticular formation
- Fibers:
  - *Afferent* - from hypothalamus through descending autonomic pathways
- Pathways:
  - Parasympathetic preganglionicl fibers -! tympanic branch of CN 9, tympanic plexus & lesser petrosal nerve -> otic ganglion
  - Efferent postganglionic -> parotid gland

3.) Sensory
- Part of solitary nuclei; sensation of taste
- Pathways:
  - Efferent fibers -> ventral nuclei of opposite thalamus and hypothalamic nuclei -> internal capsule & corona radiata -> postcentral gyrus
  - Afferent fibers -> sup ganglion of CN 9 -> spinal nuclei of CN 7
  - Carotid sinus -> tractus solitarius nucleus (dorsal motor of CN 10)
    - Function: assist in regulation of arterial BP

PATHWAYS
- Upper medulla -> post cranial fossa -> exit through jugular foramen -> upper neck with internal jugular vein & ICA -> stylopharyngeus mm -> between sup & mid constrictor mm of pharynx
  - Function: sensory of mucous membrane of pharynx & post ⅓ of tongue

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CN 10

NUCLEI

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1.) Main motor
- Location: reticular formation of medulla; formed by ambiguous nuclei
- Supply: constrictor mm of pharynx & intrinsic mm of larynx (except cricothyroid mm)
- Fibers:
  - *Corticonuclear* - both hemispheres

2.) Parasympathetic
- Location: floor of lower 4th ventricle; forms dorsal motor
- Supply: bronchi, heart, esophagus, stomach, small intestine, large intestine (distal ⅓ of transverse colon)
- Fibers:
  - *Afferent* - from hypothalamus through descending autonomic pathways

3.) Sensory nucleus
- Forms tractus solitarius nuclei
- Supply: taste sensation
- Fibers:
  - *Efferent* - ventral nuclei of opposite thalamus & hypothalamic nuclei
- Pathways:
  - Thalamic cells -> internal capsule & corona radiata -> postcentral gyrus
  - Afferent -> sup ganglion -> spinal nuclei of CN 5

PATHWAYS
- Upper medulla -> post cranial fossa -> exit in jugular foramen
  - Sup ganglion - within jugular foramen
  - Inf ganglion - below jugular foramen
- Neck within carotid sheath with internal jugular vein & ICA and common carotid
  - *Right CN 10 (post vagal trunk)* - thorax -> post root of lung -> esophagus -> diaphragm -> abdomen -> stomach, duodenum, liver, kidneys, small and large (distal ⅓ of transverse colon) intestine
  - Left CN 10 - thorax -> left aortic arch -> root of left lung -> esophagus -> diaphragm -> abdomen -> stomach, liver, upper duodenum, pancreatic head

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CN 11
- Formed by union of cranial and spinal root
- Function: movement of soft palate, pharynx, larynx, SCM & traps

CRANIAL ROOT
- Formed by ambiguous nuclei
- Fibers:
  - *Corticonuclear* - both hemisphere
  - *Efferent* - ant medulla between olive and inferior cerebellar peduncle
- Pathways:
  - Post cranial fossa -> 2 spinal root -> exit through jugular foramen -> 2 roots separate -> cranial root joins CN 10 -> pharyngeal and laryngeal branches -> mm of soft palate, pharynx, larynx

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SPINAL ROOT
- Location: ant gray column of SC (C1-5); formed by spinal nuclei
- Fibers:
  - Corticospinal - both hemispheres
- Pathways:
  - SC (between ant and post nerve roots of cervical spinal nerves) -> joins cranial root -> exit through jugular foramen -> spinal root separates cranial root -> SCM -> post triangle of neck -> trapezius

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CN 12
- Supply: intrinsic mm of tongue (styloglossus, hyoglossus, genioglossus)
- Fibers:
  - *Corticonuclear fibers* - opposite cerebral hemisphere (genioglossus)
    - Hyo and stylo are both hemispheres

NUCLEI
1.) Hypoglossal nuclei
- Location: beneath the floor of lower 4th ventricle
- Pathways:
  - Ant medulla between pyramid & olive -> post cranial fossa -> exit through hypoglossal canal -> neck between internal jugular vein & ICA -> post belly of digastric -> ICA & ECA and loop of lingual

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artery -> mylohyoid mm -> tongue mm

CLINICAL SIGNIFICANCE

1.) CN 1

1.) Anosmia - complete loss of smell
2.) Bilateral anosmia - common cold and rhinitis
3.) Unilateral anosmia - CN 1, bulb, or tract lesion
4.) Olfactory cortex lesion - complete anosmia
5.) Fx of ant cranial fossa - CN 1 tear
6.) Tumors - can cause anosmia by impingement of olfactory tract or bulb

2.) CN 2

- Causes: Expanding tumors and stroke

1.) Circumferential blindness (hysteria or optic neuritis) - spread of infection from sphenoid and ethmoid sinuses
2.) Blindness of one eye - optic nerve dissection
3.) Nasal hemianopia - partial lesion of optic chiasm on its lat side
4.) Bitemporal hemianopia - tumors in pituitary gland
5.) Contralateral homonymous hemianopia - division of optic tract or radiation or destruction of visual cortex on one side

3.) CN 3

- Causes: Diabetes, aneurysm, tumor, trauma, inflammation, vascular disease

1.) Complete lesion - cannot move eyes upward, inward, and downward
- *At rest* - eye looks laterally and downward
- *Diplopia*
- *Ptosis* - dropping of upper lids; LPS paralysis
- *Widely dilated and non-reactive to light* - sphincter pupillae paralysis & unopposed action of dilator

2.) Incomplete lesion
- *Internal ophthalmoplegia* - spared extraocular mm; affected sphincter pupillae & ciliary mm; no pupillary constriction; mydriasis
- *External ophthalmoplegia* - spared sphincter pupillae & ciliary mm; affected extraocular mm; no cardinal gaze

3.) Miosis - constricted pupil
4.) Pupil
- *If involved* - trauma, aneurysm, tumor
- *If spared* - vasculopathy

4.) CN 4

- Causes: stretching or bruising from head injuries, cavernous sinus thrombosis, aneurysm of ICA, vascular lesion of dorsal midbrain
- Signs and symptoms: diplopia when looking straight & difficulty turning eye downward & laterally

5.) CN 5

- Severe, stabbing pain over face of unknown cause; common in mandibular and maxillary (most common) division

6.) CN 6

- Causes: same as CN 4 but *vascular lesion in pons*
- Signs and symptoms: difficulty turn eye laterally, internal strabismus (unopposed medial rectus that pulls eyeball medially)

1.) Internuclear ophthalmoplegia - medial longitudinal fasciculus lesion; disconnect CN 3 nucleus from CN 6; kapag pinatingin sa gilid, naiiwan yung isang mata sa gitna
- *Bilateral* - MS, occlusive vascular disease, trauma, brainstem tumor
- *Unilateral* - basilar artery infarction

7.) CN 7

1.) CN 6 & 7 dysfunction - pons lesion
2.) CN 7 & 8 dysfunction - internal acoustic meatus lesion
3.) Excessively sensitive in 1 ear - stapedius lesion
4.) Loss of taste in ant ⅓ of tongue - proximal damage
5.) Swelling of parotid gland with impaired function of CN 7 - malignancy of parotid gland
6.) Deep lacerations - may damage CN 7
7.) UMN lesion - mm of lower face is paralyzed
8.) LMN lesion or CN 7 lesion - all mm affected

of face will be paralyzed

- Lower lid drop, sagging angle of mouth, tears over lower lid, saliva dribbles from corners, unable to close eye. unable to expose teeth fully on affected side

9.) Hemiplegia - emotional movements are preserved
10.) Bell's palsy - dysfunction of CN 7; unilateral; LMN type of paralysis

8.) CN 8

1.) Disturbance of vestibular nerve:
- *Giddiness (vertigo)* - severe dizziness; rotatory dizziness
- *Nystagmus* - ataxia of ocular mm

2.) Vestibular nystagmus - uncontrollable rhythmic oscillation of eyes; disturbance in reflex control of extraocular mm

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1.) Disturbance of cochlear nerve:
- *Deafness* - sensorineural (inner ear) & conductive (mid and outer ear)
- *Tinnitus* - flat line sound

2.) Loss of hearing
- May be due to:
  - Defective auditory conducting mechanism of middle ear
  - Receptor cell damage in spiral organ of Corti
  - Lesion of cochlear nerve, central auditory pathways, temporal cortex

3.) Inner ear lesion - Meniere's disease (severe vertigo) & acute labyrinthitis
4.) Cochlear nerve lesion - tumor (acoustic neuroma) & trauma
5.) Lesion of CNS - midbrain tumors and MS

9.) CN 9
- Isolated lesions are rare and is usually related with CN 10

10.) CN 10
- Lesions involving CN 10 also involve CN 9, 11, & 12

1.) Hoarseness - vagal nerve palsy; all mm

11.) CN 11
- Causes: tumors, tab or gunshot wounds

1.) CN 11 lesion - SCM and trapezium paralysis
2.) SCM atrophy - turn the head to opposite side
3.) Trapezius atrophy - shoulder droop on ipsilateral side; weakness & difficulty raising arm on horizontal plance

12.) CN 12
- Causes: tumor, demyelinating disease, syringomyelia, vascular accidents, stab & gunshot wounds

1.) LMN lesions - tongue will deviate towards the ipsilateral side, is smaller due to atrophy, and fasciculation may accompany with atrophy